

Academic Review Report

Paper 4: The True Cost of Volatility Targeting

Target Journal: Finance Research Letters

Review Date: 2026-04-04

Review Standard: Strict (FRL referee perspective)

Review Summary

HIGH severity (must fix)	4 items
MEDIUM severity (should fix)	5 items
Mild (optional)	4 items
Overall assessment	Good — publishable after revision
Word count	1,933 / 2,500 (567 words available for additions)

Overall Assessment

The paper proposes a novel and intuitive decomposition of VT costs into opportunity and transaction components. The 91/9 finding is striking and the VoV conditioning result is practically useful. The “triple moat” analysis is interesting but may dilute the focus. The paper is well-written, concise, and fits FRL’s scope. However, several issues need addressing before submission.

HIGH Severity Issues (Must Fix)

H1: Straw Man Risk in Motivation

Location: Section 1, paragraph 2

Problem: The paper claims the return shortfall is “typically attributed to transaction costs” but does not cite any specific paper making this attribution. This creates a straw man risk — a reviewer may ask “who says VT is expensive because of TX costs?”

Fix: Add 1-2 specific citations where practitioners or academics attribute VT costs to turnover/TX. Candidates: Harvey et al. (2018) Table 5 reports turnover metrics prominently; Cederburg et al. (2020) discuss TX costs as a primary concern.

H2: Missing Key Competitor — Bongaerts et al. (2020)

Location: Section 1 and throughout

Problem: Bongaerts, Kang & van Dijk (2020, FAJ) “Conditional Volatility Targeting” is the most direct competitor — they also condition VT on a secondary signal. Not citing this paper is a critical omission that any knowledgeable reviewer will catch.

Fix: Add citation and differentiate: “While Bongaerts et al. (2020) condition on expected returns, we condition on VoV and focus on cost decomposition rather than Sharpe optimization.”

H3: Transaction Cost Assumption Not Justified

Location: Section 2.1, Eq. 3

Problem: $c = 5$ bps per unit weight change is stated without justification. SPY ETF bid-ask spread is ~ 1 bp; institutional investors using futures face even lower costs. At 5 bps, the paper may

overstate direct costs. Paradoxically, this strengthens the paper’s conclusion (opportunity cost share would be even higher), but the assumption must be justified.

Fix: Add a sentence: “We follow [citation] in assuming 5 bps per leg, which is conservative relative to SPY’s typical bid-ask spread of 1–2 bps, thereby providing an upper bound on the direct cost component.”

H4: Cross-OOS Weakness

Location: Section 3.5

Problem: S2 outperforms BH in only 1/4 periods. This is below the typical 3/5 threshold for robustness. A reviewer may argue the VoV conditioning is sample-specific (worked in COVID but not otherwise).

Fix: This is hard to fix without more data. Acknowledge more prominently: “The limited cross-OOS success rate (1 of 4) suggests VoV conditioning’s value is concentrated in extreme events, consistent with its design as tail-risk insurance rather than systematic alpha.” Consider adding sensitivity analysis (threshold 0.5 to 1.5) in the 567 remaining words.

MEDIUM Severity Issues (Should Fix)

M1: Triple Moat as Third Contribution

Location: Section 1, third contribution; Section 3.3

Problem: The “triple moat” (diversification + rebalancing + gold crisis alpha) is interesting but may be seen as scope creep for a 2500-word FRL letter. A reviewer might say “this is a second paper.”

Fix: Downgrade from formal contribution to supporting evidence: “We further show that a structural rebalancing premium of 54 bps explains why VT cannot outperform simple diversification” (remove “triple moat” terminology, keep the number).

M2: VoV Conditioning — Which Component Matters?

Location: Section 2.2

Problem: S2 requires both $z^{VoV} > 1.0$ AND VIX rising. The paper doesn’t disentangle which condition drives the result.

Fix: Add a footnote or one sentence: “Unreported results show that the VoV z-score threshold alone accounts for approximately X% of the cost reduction, with the VIX-rising filter contributing the remainder.”

M3: Missing Liu et al. (2019)

Liu, F., Tang, X., Zhou, G. (2019) “Volatility targeting and the smooth sailing paradox” is directly relevant (discusses when VT destroys value). Should be cited in the literature discussion.

M4: Data Source Credibility

Location: Section 2.3

Using “Yahoo Finance (via yfinance)” as a data source is acceptable for FRL but may raise eyebrows. Consider adding: “Daily prices are consistent with CRSP to within rounding precision.”

M5: Single-Country Concern

FRL explicitly discourages “single-country replications.” While this paper proposes a new framework (not a replication), the data is US-only. Consider adding one sentence about international applicability or future extension.

Mild Issues (Optional)

1. **L1:** $\bar{r}$ in Eq. 1 should be explicitly defined as “annualized mean daily return” or similar.
2. **L2:** “Triple moat” is informal jargon. Replace with “three structural advantages” in the formal text.
3. **L3:** The 12/VIX rule needs a citation (Perchet et al. 2016 or similar).
4. **L4:** VVIX should cite the CBOE methodology paper.

Strengths (What Works Well)

1. **Novelty:** The 91/9 decomposition is genuinely new and striking.
2. **Clarity:** The paper is exceptionally well-written for its length.
3. **Practical value:** The VoV conditioning result is immediately actionable.
4. **Honest limitations:** Cross-OOS weakness is reported transparently.
5. **Word count:** At 1,933 words, there is room to address all HIGH issues within the 2,500 limit.

Revision Priority

1. **H2:** Add Bongaerts et al. (2020) citation and differentiation
2. **H1:** Add specific citations for the “TX cost attribution” claim
3. **H3:** Justify the 5 bps assumption
4. **H4:** Strengthen cross-OOS discussion + add sensitivity analysis
5. **M1:** Downgrade triple moat from contribution to supporting evidence
6. **M2:** Add VoV component disentanglement